No Man's Land

A custom SystemVerilog GPU and game on the DE1-SoC

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CSEE 4840 — Embedded System Design — Spring 2026

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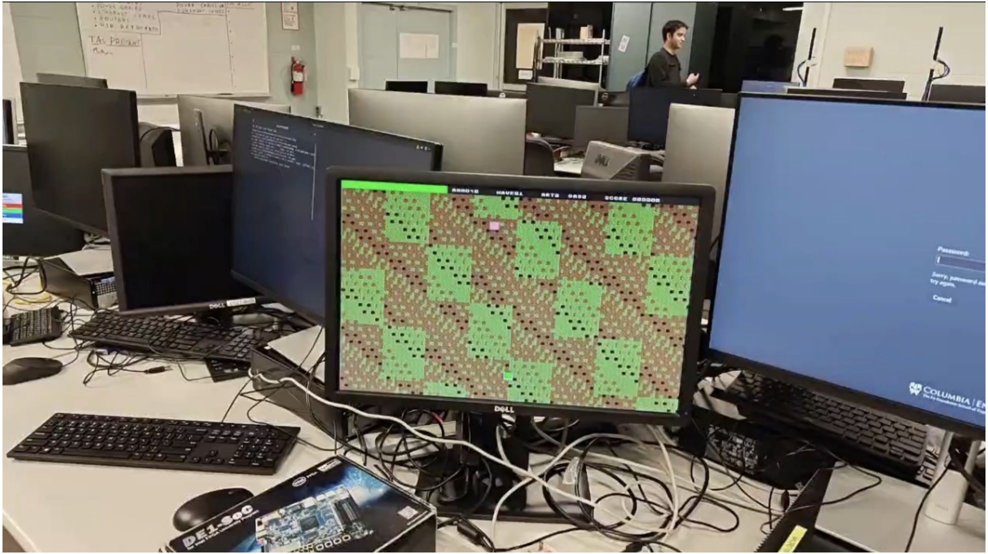
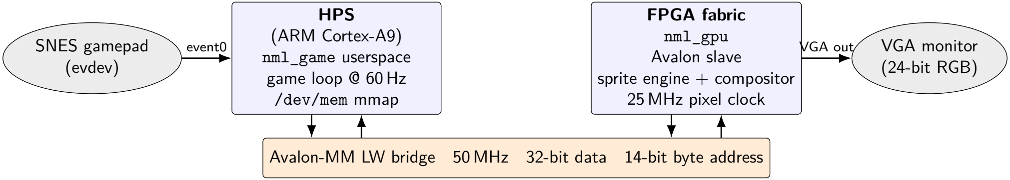
The system, running

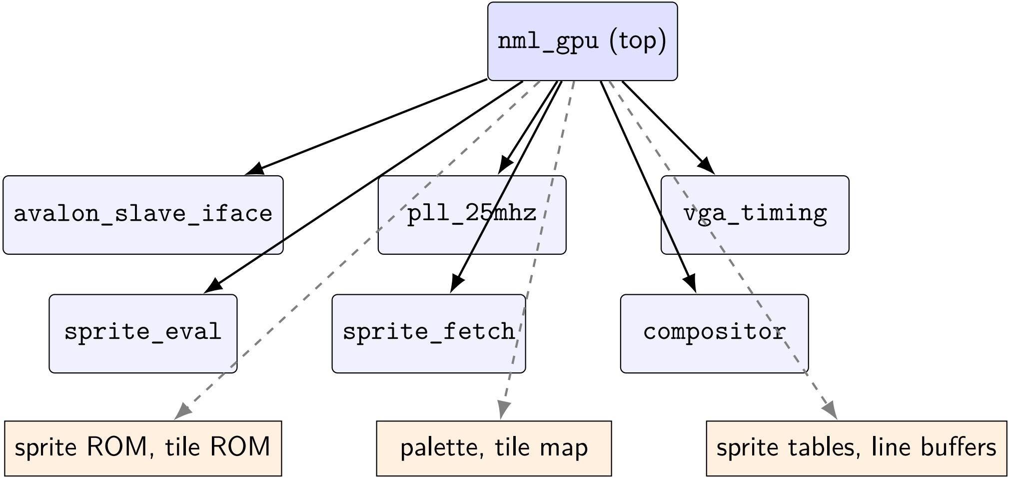


System block



HPS writes shadow state; FPGA latches at VSYNC; pixels always stream.

What's inside `nml_gpu`

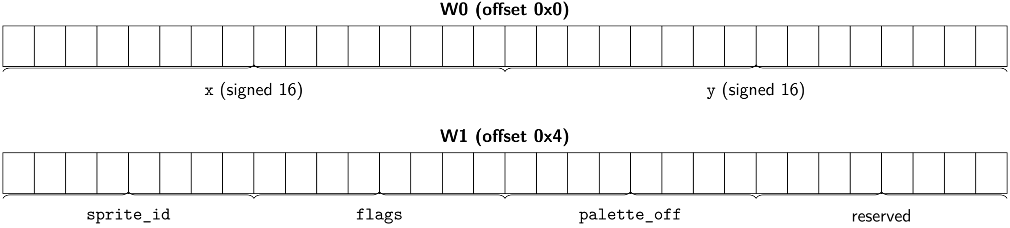


One 50 MHz domain (Avalon) + one 25 MHz domain (pixel). All RAMs/ROMs inferred as M10K.

Avalon register map — 16 KB window at 0xFF200000

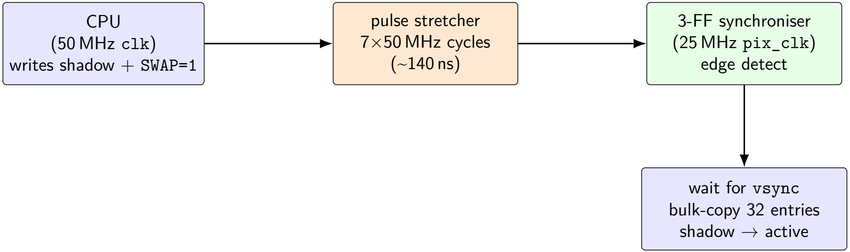
Offset	Name	Layout
`0x0000`	`CTRL`	`[0]`=ENABLE, `[1]`=SWAP (auto-clear), `[2]`=HUD_ON
`0x0004`	`STATUS`	`[0]`=VBLANK, `[1]`=SWAP_PENDING
`0x0008`	`BG_SCROLL`	`[15:0]`=scroll_x, `[31:16]`=scroll_y (signed)
`0x000C`	`IRQ_MASK`	`[0]`=VBLANK IRQ enable
`0x0010`	`PLAYER_POS`	`[15:0]`=px, `[31:16]`=py
`0x0014`	`PLAYER_STATS`	`[7:0]`=hp, `[15:8]`=wave (BCD), `[31:16]`=level
`0x0018`	`SCORE`	`[23:0]`=6-digit BCD score
`0x001C`	`KILL_COUNT`	32-bit kills
`0x0020`	`HUD_AUX`	`[7:0]`=ammo BCD, `[11:8]`=art, `[15:12]`=gas
`0x0100-0x01FF`	**Sprite table**	32 entries × 8 B
`0x0400-0x07FF`	**Palette**	256 entries × 4 B (RGB888)
`0x1000-0x22BF`	**Tile map**	80 cols × 60 rows, 1 byte/tile

Sprite-table entry — 8 bytes / 64 bits



flags byte: [2] hflip [3] vflip [6:4] priority (0 = highest) [7] ACTIVE

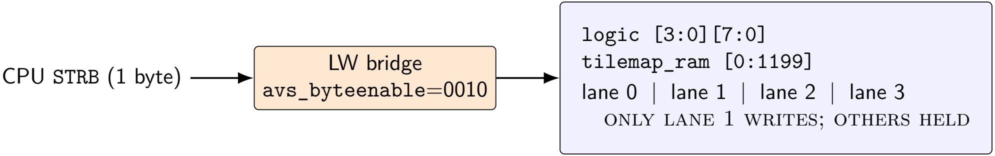
Frame commit — shadow → active at VSYNC



CPU then polls STATUS.SWAP_PENDING;
clears one cycle after the bulk copy.

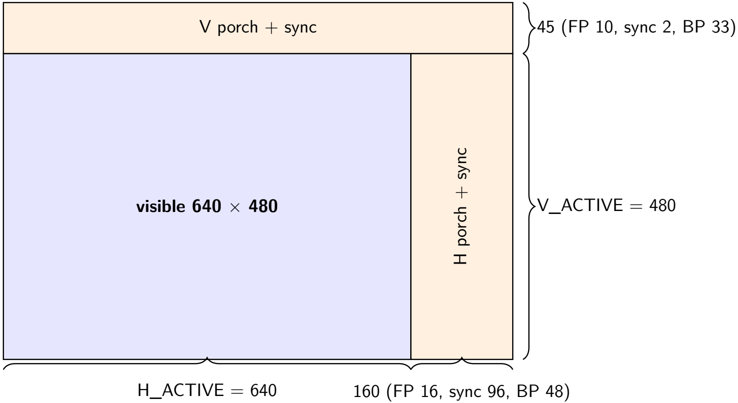
One 20 ns Avalon pulse would slip past the 40 ns pixel clock. Widen to 140 ns and the CDC is deterministic.

Tile-map byte writes — packed `[3:0][7:0]`



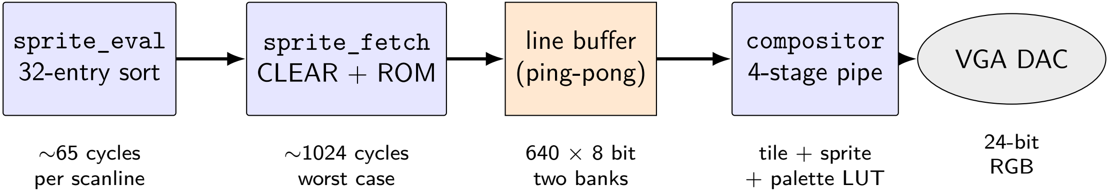
Why packed? Byte-organised storage forced bridge RMW to clobber 3 lanes per word. Packed inference uses M10K `byteena_a`: one lane updates, the other three are untouched.

VGA timing — 640×480 @ 60 Hz, 25 MHz dot



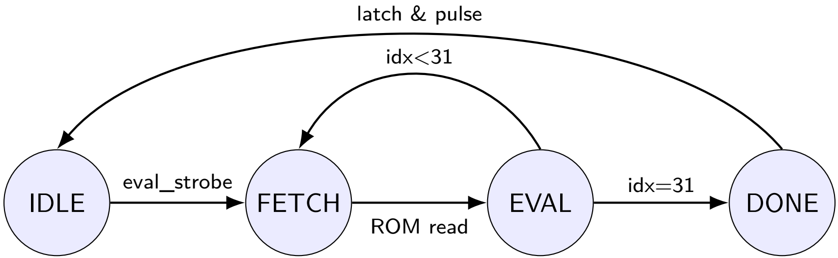
One frame = 800 × 525 dot cycles. HBLANK is the 160-cycle window we use to evaluate sprites for the next scanline.

Pixel pipeline — sprite engine + compositor



Eval runs inside HBLANK (160 cycles). Fetch spans HBLANK + visible (800 cycles).
Line buffers are double-buffered, so fetch lags read by one scanline. Compositor never stalls.

`sprite_eval` — per-scanline priority sort



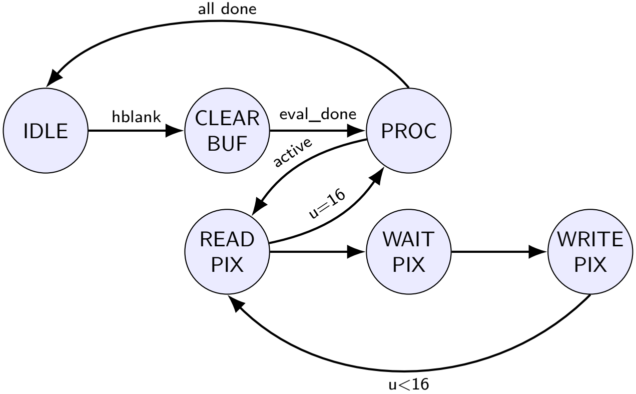
Per sprite (1 cycle in EVAL):

`intersects = active && y≤next_scanline < y+16`

insertion-sort into 8 priority-ordered slots (all 8 compares in parallel)

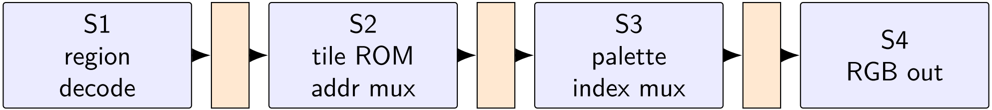
Total: 32 fetch+eval cycles + 1 DONE = ∼65 cycles. Fits easily in 160-cycle HBLANK.

`sprite_fetch` — ROM read + line-buffer fill



Worst-case cycle budget per scanline: CLEAR_BUF (640) + 8 sprites × 48 (384) =∼1024 cycles. HBLANK alone is 160; HBLANK + visible is **800**. Fetch lives one line behind the read, so the ping-pong line buffers absorb the latency.

Compositor — 4-stage pipeline



HUD predicates,
HUD tile ID,
BCD digit

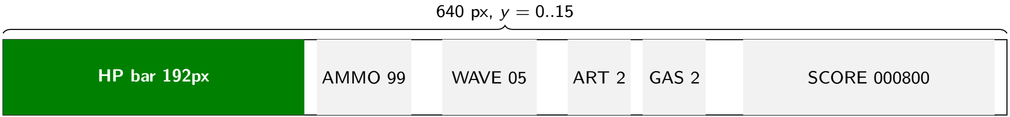
bg tile ROM
or HUD glyph

sprite px if $\neq 0$
else tile px

HUD layer
wins on top

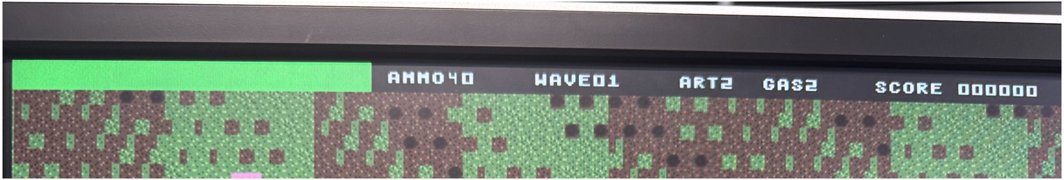
Latches between stages absorb the 1-cycle latency of tile ROM and palette M10K reads. Never stalls.

HUD strip — top 16 pixels of every frame



HP bar: 2 px per HP, clamped to 192. Glyphs: 8×8 from tile ROM, slots 16–41 (A–Z), 48–57 (0–9). Scores, ammo, wave packed BCD in registers.

HUD — on the monitor



Memory inventory (all inferred M10K)

Memory	Shape	Init	Bits
sprite ROM	$16{,}384 \times 8$	`$readmemh`	131,072
tile ROM	$4{,}096 \times 8$	`$readmemh`	32,768
tile map	$1{,}200 \times 32$ (packed 4×8)	blank (HPS writes)	38,400
palette	256×24	`$readmemh` + HPS overrides	6,144
sprite table act.	32×64	via VSYNC bulk copy	2,048
sprite table shd.	32×64	HPS writes	2,048
line buffer A	640×8	cleared per HBLANK	5,120
line buffer B	640×8	cleared per HBLANK	5,120
Total			**86,144 bits**

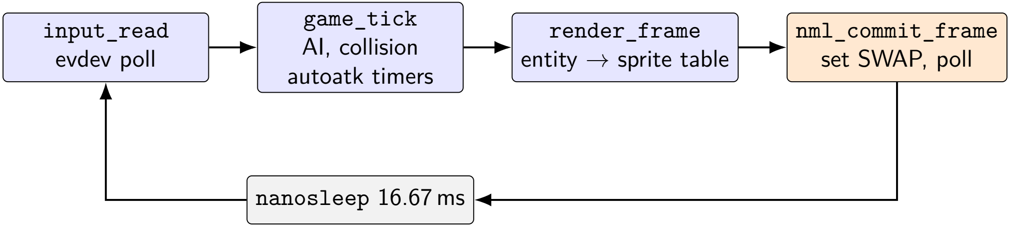
14 M10K blocks out of 397.

Resource utilisation — Cyclone V 5CSEMA5F31C6

Resource	Used	Available	%
ALMs	205	32,070	$< 1\%$
Registers	365	—	—
Pins	54	457	12%
Block memory bits	86,144	4,065,280	2%
M10K blocks	14	397	4%
DSP blocks	0	87	0%

Compile time: 63 s wall clock (Analysis 13 s, Fitter 30 s, Asm 12 s, STA 8 s).

Software architecture — 60 Hz game loop on the ARM



`game_t` holds a 64-entity pool. State machine: PLAYING ↔ LEVELUP → GAMEOVER.
Mortar is timer-driven; artillery + gas fire from L/R shoulders.

What actually works

640×480 @ 60 Hz • tile-mapped battlefield • 32 hardware sprites • 256-entry palette
HUD: HP bar, AMMO, WAVE, ART, GAS, SCORE (BCD)
SNES gamepad via evdev; 60 Hz HPS game loop; frame-commit at VSYNC
Mortar, artillery, gas clouds, wave spawner, level-up, ammo drops, game-over